

Advanced Visualizations

```
library(gganimate)
```

Warning: package 'gganimate' was built under R version 4.5.1

Loading required package: ggplot2

```
library(ggplot2)  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(readr)  
library(tidyr)  
library(gifski)
```

Warning: package 'gifski' was built under R version 4.5.1

```
library(leaflet)
```

Warning: package 'leaflet' was built under R version 4.5.1

```
library(sf)
```

Warning: package 'sf' was built under R version 4.5.1

Linking to GEOS 3.13.1, GDAL 3.11.0, PROJ 9.6.0; sf_use_s2() is TRUE

```
library(RColorBrewer)  
library(stringr)  
library(ragg)
```

Warning: package 'ragg' was built under R version 4.5.1

- Using PSA OpenStat (2013-2022) data, create an animated bar chart showing the total number of live births per year. Differentiate Male vs. Female births using `fill`

```
birth_data <- read_csv("PSA_both.t.csv", skip = 2)
```

Rows: 17 Columns: 21

— Column specification —

Delimiter: ","

chr (1): Region, Province/Highly Urbanized City

dbl (20): 2013 Male, 2013 Female, 2014 Male, 2014 Female, 2015 Male, 2015 Fe...

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
birth_totals <- birth_data %>%
  select(-1) %>%
  summarise_all(sum, na.rm = TRUE) %>%
  pivot_longer(
    cols = everything(),
    names_to = "Year_Gender",
    values_to = "Births"
  ) %>%
  separate(Year_Gender, into = c("Year", "Gender"), sep = " ") %>%
  mutate(
    Year = as.numeric(Year),
    Births = as.numeric(Births)
  )

head(birth_totals)
```

A tibble: 6 × 3

	Year	Gender	Births
	<dbl>	<chr>	<dbl>
1	2013	Male	917072
2	2013	Female	844451
3	2014	Male	912425
4	2014	Female	836357
5	2015	Male	910804
6	2015	Female	833823

```
p <- birth_totals %>%
  ggplot(aes(x = Year, y = Births, fill = Gender)) +
  geom_col(position = "dodge", width = 0.7) +
  scale_fill_manual(values = c("Male" = "steelblue", "Female" = "coral")) +
  labs(
    title = "Total Live Births by Year and Gender (Philippines)",
    x = "Year",
    y = "Number of Live Births"
  ) +
```

```

theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, size = 14),
  axis.text.x = element_text(angle = 45, hjust = 1)
)
library(plotly)

```

Warning: package 'plotly' was built under R version 4.5.1

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

The following object is masked from 'package:stats':

filter

The following object is masked from 'package:graphics':

layout

```

animated_plot <- plot_ly(
  data = birth_totals,
  x = ~Gender,
  y = ~Births,
  color = ~Gender,
  colors = c("steelblue", "coral"),
  frame = ~Year,
  type = "bar"
) %>%
  layout(
    title = "Animated Live Births by Gender and Year (Philippines)",
    xaxis = list(title = "Gender"),
    yaxis = list(title = "Number of Live Births")
  ) %>%
  animation_opts(
    frame = 1000,
    transition = 500,
    redraw = FALSE
  )

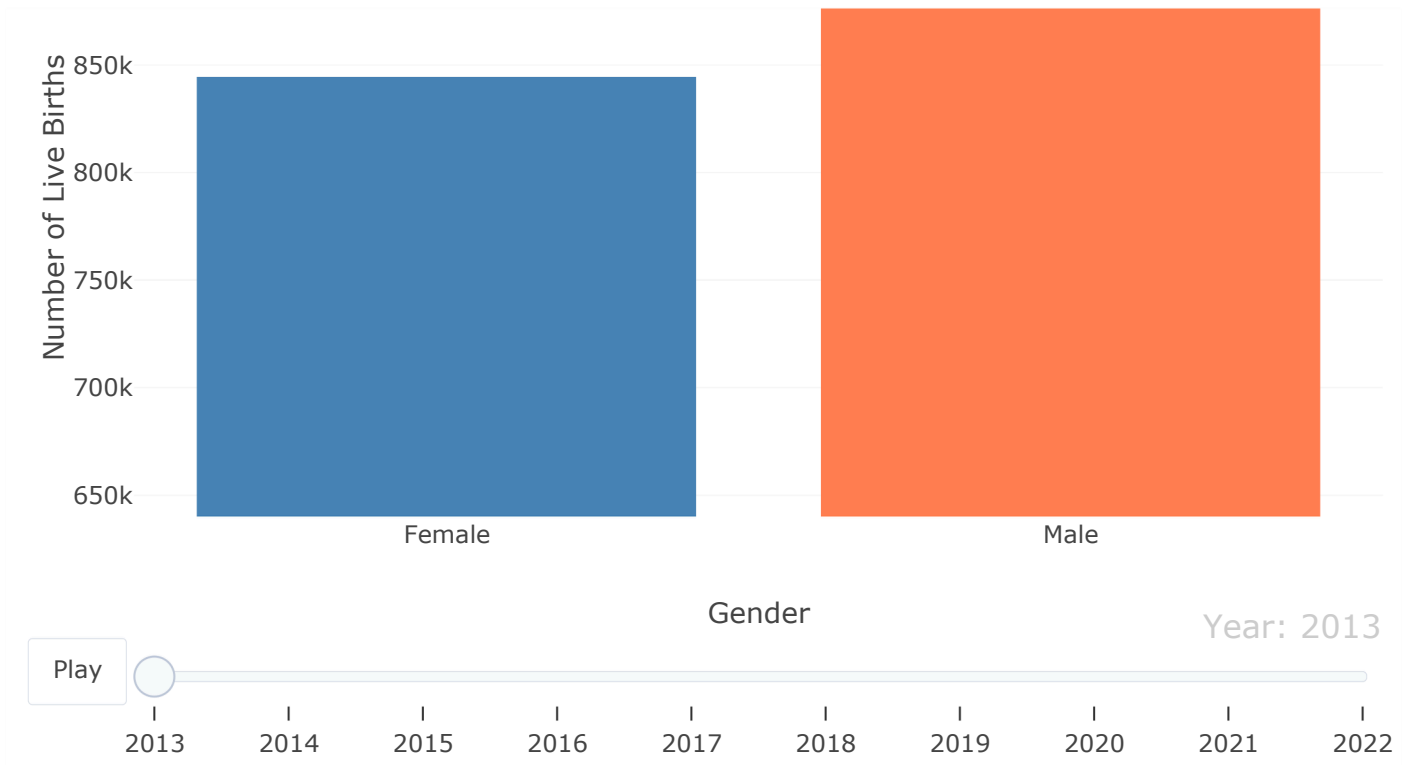
animated_plot

```

Animated Live Births by Gender and Year (Philippines)

900k





- Create an animated bar chart that compares the number of live births per region over time (2013-2022). Use `transition_states(year, wrap = FALSE)` to animate regional birth counts changing over time. Determine the region with the highest live births in each frame.

```
birth_data <- read_csv("PSA_both.t.csv", skip = 2)
```

Rows: 17 Columns: 21

— Column specification —

Delimiter: ","

chr (1): Region, Province/Highly Urbanized City

dbl (20): 2013 Male, 2013 Female, 2014 Male, 2014 Female, 2015 Male, 2015 Fe...

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
birth_regional <- birth_data %>%
  mutate(
    `2013_Total` = `2013 Male` + `2013 Female`,
    `2014_Total` = `2014 Male` + `2014 Female`,
    `2015_Total` = `2015 Male` + `2015 Female`,
    `2016_Total` = `2016 Male` + `2016 Female`,
    `2017_Total` = `2017 Male` + `2017 Female`,
    `2018_Total` = `2018 Male` + `2018 Female`,
    `2019_Total` = `2019 Male` + `2019 Female`,
    `2020_Total` = `2020 Male` + `2020 Female`,
    `2021_Total` = `2021 Male` + `2021 Female`,
    `2022_Total` = `2022 Male` + `2022 Female`
  ) %>%
  select(1, contains("Total")) %>%
```

```

pivot_longer(
  cols = -1,
  names_to = "Year",
  values_to = "Births"
)%>%
mutate(
  Year = as.numeric(gsub("_Total", "", Year)),
  Region = `Region, Province/Highly Urbanized City`
) %>%
select(Region, Year, Births) %>%
group_by(Year) %>%
mutate(
  Rank = rank(-Births, ties.method = "first"),
  IsHighest = Rank == 1
) %>%
ungroup()

birth_regional_bar <- ggplot(birth_regional, aes(x = reorder(Region, Births), y = Births, fill = I
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("FALSE" = "lightblue", "TRUE" = "darkred")) +
  labs(
    title = "Regional Live Births: {closest_state}",
    x = "Region",
    y = "Number of Live Births",
    fill = "Highest Region"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(size = 16, hjust = 0.5)
  ) +
  transition_states(Year, transition_length = 2, state_length = 1, wrap = FALSE)

# Define output file BEFORE using it
anim_file2 <- "regional_births.gif"

# Load birth data again
birth_data <- read_csv("PSA_both.t.csv", skip = 2)

```

Rows: 17 Columns: 21

— Column specification —————

Delimiter: ","

chr (1): Region, Province/Highly Urbanized City

dbl (20): 2013 Male, 2013 Female, 2014 Male, 2014 Female, 2015 Male, 2015 Fe...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

# Data wrangling
birth_regional <- birth_data %>%
  mutate(
    `2013_Total` = `2013 Male` + `2013 Female`,
    `2014_Total` = `2014 Male` + `2014 Female`,
    `2015_Total` = `2015 Male` + `2015 Female`,
    `2016_Total` = `2016 Male` + `2016 Female`,
    `2017_Total` = `2017 Male` + `2017 Female`,
    `2018_Total` = `2018 Male` + `2018 Female`,
    `2019_Total` = `2019 Male` + `2019 Female`,
    `2020_Total` = `2020 Male` + `2020 Female`,
    `2021_Total` = `2021 Male` + `2021 Female`,
    `2022_Total` = `2022 Male` + `2022 Female`
  ) %>%
  select(1, contains("Total")) %>%
  pivot_longer(
    cols = -1,
    names_to = "Year",
    values_to = "Births"
  ) %>%
  mutate(
    Year = as.numeric(gsub("_Total", "", Year)),
    Region = `Region, Province/Highly Urbanized City`
  ) %>%
  select(Region, Year, Births) %>%
  group_by(Year) %>%
  mutate(
    Rank = rank(-Births, ties.method = "first"),
    IsHighest = Rank == 1
  ) %>%
  ungroup()

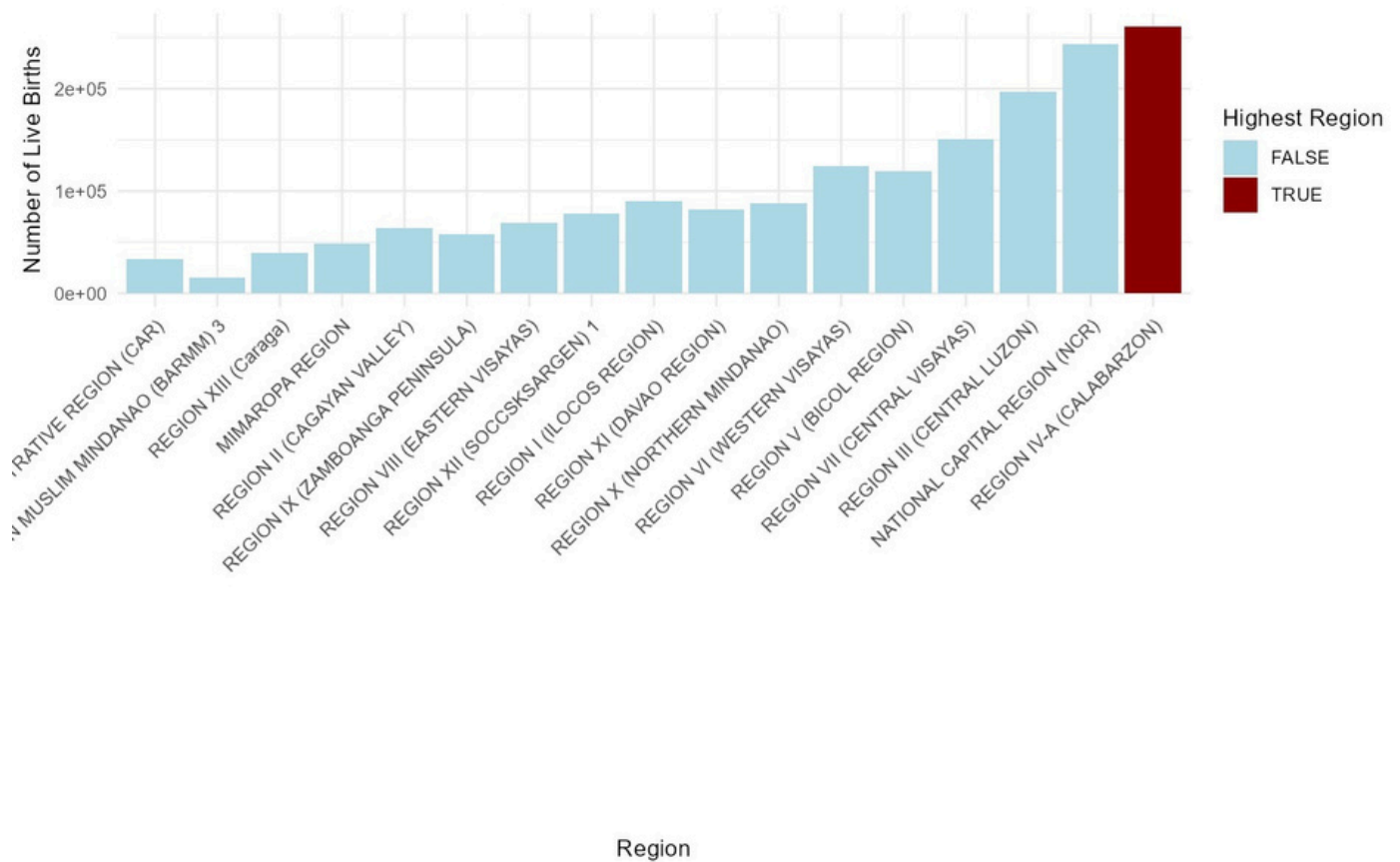
# ggplot object
birth_regional_bar <- ggplot(birth_regional, aes(x = reorder(Region, Births), y = Births, fill = I
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("FALSE" = "lightblue", "TRUE" = "darkred")) +
  labs(
    title = "Regional Live Births: {closest_state}",
    x = "Region",
    y = "Number of Live Births",
    fill = "Highest Region"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(size = 16, hjust = 0.5)
  ) +
  transition_states(Year, transition_length = 2, state_length = 1, wrap = FALSE)

# Animate and save to file

```

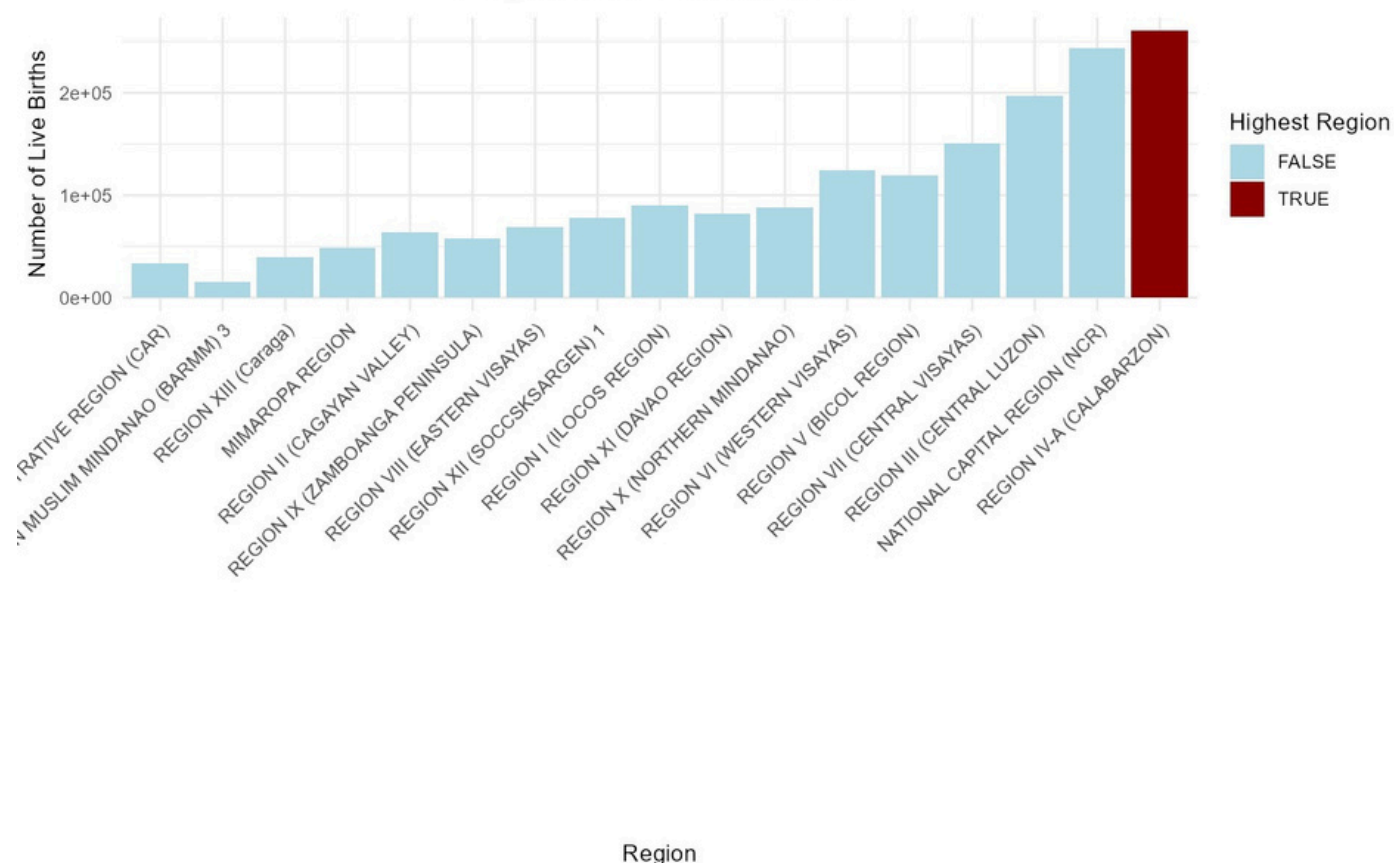
```
animate(
  birth_regional_bar,
  duration = 10,
  fps = 10,
  width = 9,
  height = 6,
  res = 150,
  renderer = gifski_renderer(anim_file2),
  device = "ragg_png"    # use string name, not the function
)
```

Regional Live Births: 2013



```
# Include gif in output
knitr::include_graphics(anim_file2)
```

Regional Live Births: 2013



- Select 3-5 regions and visualize their live birth trends (2013-2022) using an animated time series plot.

```
birth_data <- read_csv("PSA_both.t.csv", skip = 2)
```

Rows: 17 Columns: 21

— Column specification

Delimiter: ",",

chr (1): Region, Province/Highly Urbanized City

dbl (20): 2013 Male, 2013 Female, 2014 Male, 2014 Female, 2015 Male, 2015 Fe...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
selected_regions <- c("NATIONAL CAPITAL REGION (NCR)",
                     "CORDILLERA ADMINISTRATIVE REGION (CAR)",
                     "MIMAROPA REGION")
```

```
birth_three_regions <- birth_data %>%
  filter(`Region, Province/Highly Urbanized City` %in% selected_regions) %>%
  mutate(
    `2013` = `2013 Male` + `2013 Female`,
    `2014` = `2014 Male` + `2014 Female`,
    `2015` = `2015 Male` + `2015 Female`,
    `2016` = `2016 Male` + `2016 Female`,
```



```

`2017` = `2017 Male` + `2017 Female`,
`2018` = `2018 Male` + `2018 Female`,
`2019` = `2019 Male` + `2019 Female`,
`2020` = `2020 Male` + `2020 Female`,
`2021` = `2021 Male` + `2021 Female`,
`2022` = `2022 Male` + `2022 Female`
) %>%
select(1, `2013`:`2022`) %>%
pivot_longer(
  cols = -1,
  names_to = "Year",
  values_to = "Births"
) %>%
mutate(
  Year = as.numeric(Year),
  Region = case_when(
    `Region, Province/Highly Urbanized City` == "NATIONAL CAPITAL REGION (NCR)" ~ "NCR",
    `Region, Province/Highly Urbanized City` == "CORDILLERA ADMINISTRATIVE REGION (CAR)" ~ "CAR",
    `Region, Province/Highly Urbanized City` == "MIMAROPA REGION" ~ "MIMAROPA"
  )
) %>%
select(Region, Year, Births)

multi_line_plot <- birth_three_regions %>%
  ggplot(aes(x = Year, y = Births, color = Region)) +
  geom_line(size = 1.8, alpha = 0.7) +
  geom_point(size = 4, alpha = 0.9) +
  scale_color_manual(values = c("NCR" = "darkred", "CAR" = "navy", "MIMAROPA" = "forestgreen")) +
  labs(
    title = "Live Birth Trends: Three Key Regions",
    subtitle = "Animated comparison (2013-2022) - Year: {frame_along}",
    x = "Year",
    y = "Live Births",
    color = "Region"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    legend.position = "bottom",
    legend.title = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  ) +
  scale_x_continuous(breaks = 2013:2022) +
  scale_y_continuous(labels = scales::comma_format()) +
  transition_reveal(Year)

```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.

```
# Define the output filename BEFORE animate()
anim_file3 <- "three_regions_timeseries.gif"

# Data processing for the 3 selected regions
birth_data <- read_csv("PSA_both.t.csv", skip = 2)
```

Rows: 17 Columns: 21

— Column specification —

Delimiter: ",",

chr (1): Region, Province/Highly Urbanized City

dbl (20): 2013 Male, 2013 Female, 2014 Male, 2014 Female, 2015 Male, 2015 Fe...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
selected_regions <- c(
  "NATIONAL CAPITAL REGION (NCR)",
  "CORDILLERA ADMINISTRATIVE REGION (CAR)",
  "MIMAROPA REGION"
)

birth_three_regions <- birth_data %>%
  filter(`Region, Province/Highly Urbanized City` %in% selected_regions) %>%
  mutate(
    `2013` = `2013 Male` + `2013 Female`,
    `2014` = `2014 Male` + `2014 Female`,
    `2015` = `2015 Male` + `2015 Female`,
    `2016` = `2016 Male` + `2016 Female`,
    `2017` = `2017 Male` + `2017 Female`,
    `2018` = `2018 Male` + `2018 Female`,
    `2019` = `2019 Male` + `2019 Female`,
    `2020` = `2020 Male` + `2020 Female`,
    `2021` = `2021 Male` + `2021 Female`,
    `2022` = `2022 Male` + `2022 Female`
  ) %>%
  select(1, `2013`:`2022`) %>%
  pivot_longer(
    cols = -1,
    names_to = "Year",
    values_to = "Births"
  ) %>%
  mutate(
    Year = as.numeric(Year),
    Region = case_when(
      `Region, Province/Highly Urbanized City` == "NATIONAL CAPITAL REGION (NCR)" ~ "NCR",
      `Region, Province/Highly Urbanized City` == "CORDILLERA ADMINISTRATIVE REGION (CAR)" ~ "CAR",
      `Region, Province/Highly Urbanized City` == "MIMAROPA REGION" ~ "MIMAROPA"
    )
  ) %>%
  select(Region, Year, Births)
```

```

# ggplot object for the animated line chart
multi_line_plot <- birth_three_regions %>%
  ggplot(aes(x = Year, y = Births, color = Region)) +
  geom_line(size = 1.8, alpha = 0.7) +
  geom_point(size = 4, alpha = 0.9) +
  scale_color_manual(values = c("NCR" = "darkred", "CAR" = "navy", "MIMAROPA" = "forestgreen")) +
  labs(
    title = "Live Birth Trends: Three Key Regions",
    subtitle = "Animated comparison (2013-2022) – Year: {frame_along}",
    x = "Year",
    y = "Live Births",
    color = "Region"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    legend.position = "bottom",
    legend.title = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  ) +
  scale_x_continuous(breaks = 2013:2022) +
  scale_y_continuous(labels = scales::comma_format()) +
  transition_reveal(Year)

# Animate the plot and save as a .gif
# Animate and save time series for three regions (fix: device = "ragg_png")
animate(
  multi_line_plot,
  duration = 12,
  fps = 10,
  width = 9,
  height = 6,
  res = 150,
  renderer = gifski_renderer(anim_file3),
  device = "ragg_png" # use string name
)

```

`geom\_line()`: Each group consists of only one observation.

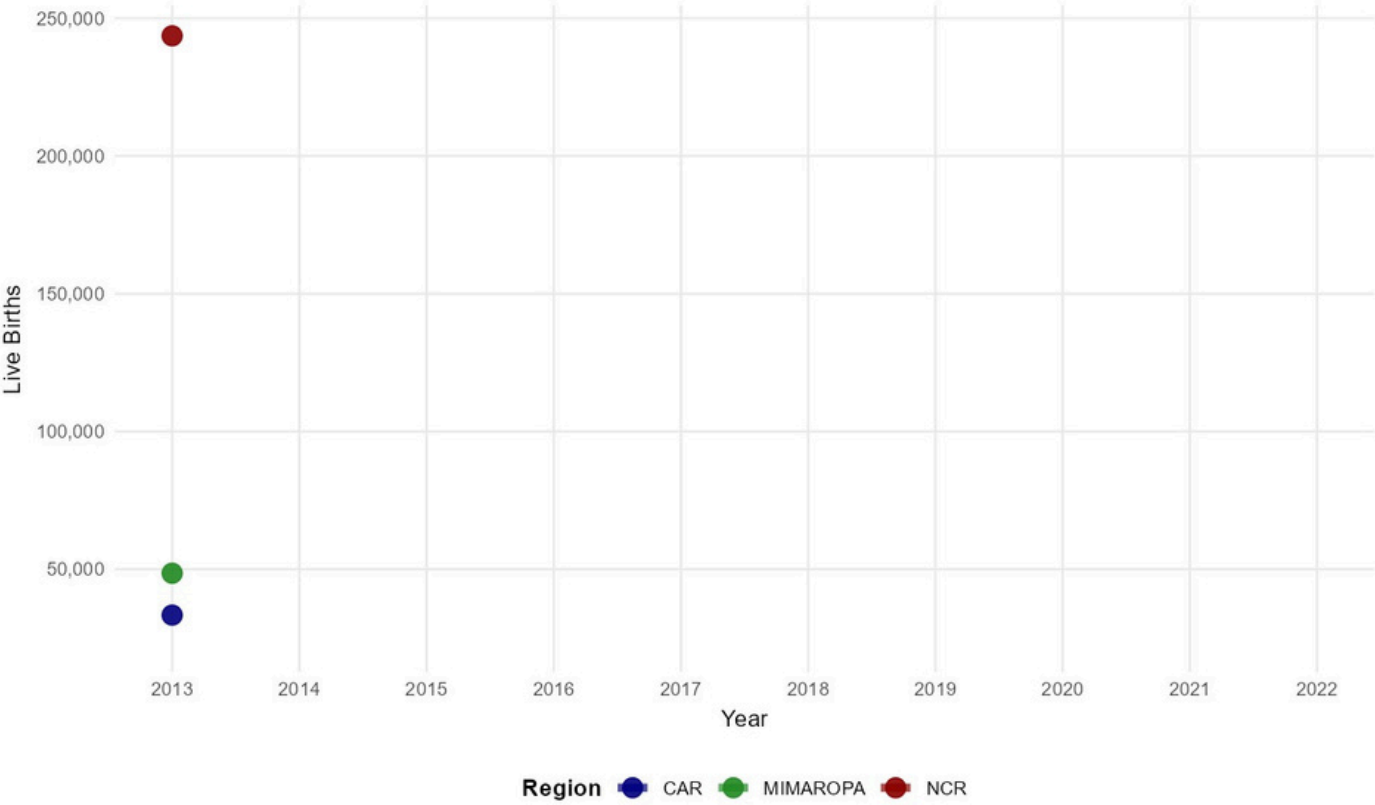
i Do you need to adjust the group aesthetic?

`geom\_line()`: Each group consists of only one observation.

i Do you need to adjust the group aesthetic?

Live Birth Trends: Three Key Regions

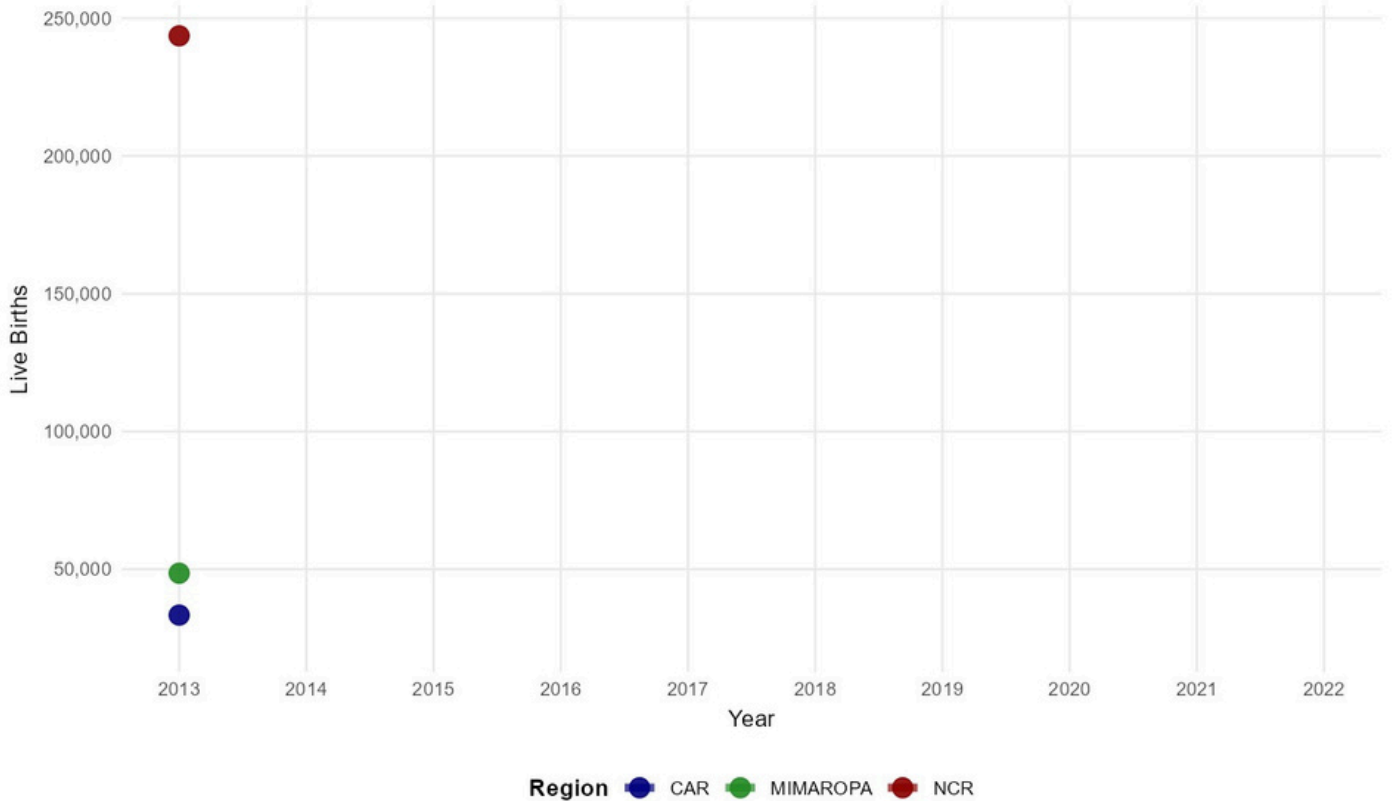
Animated comparison (2013–2022) — Year: 2013



```
# Display the GIF in Quarto output
knitr::include_graphics(anim_file3)
```

Live Birth Trends: Three Key Regions

Animated comparison (2013–2022) — Year: 2013



- Using PSA OpenStat (2013-2022) data and a Philippines regional shapefile, create a static choropleth map of the recent number of live births per region. Color the regions based on birth count

```
# Read the shapefile
ph_regions <- st_read("philippines_Regions_level_1.shp")
```

```
Reading layer `philippines_Regions_level_1' from data source
`C:\Users\Kimberly\R - Ylrebmik\Advanced Visualizations\philippines_Regions_level_1.shp'
using driver `ESRI Shapefile'
Simple feature collection with 10 features and 7 fields
Geometry type: MULTIPOLYGON
Dimension: XY
Bounding box: xmin: 114.2779 ymin: 4.587294 xmax: 125.0077 ymax: 21.12189
Geodetic CRS: WGS 84
```

```
# Print region names in shapefile for debugging
cat("Shapefile region names:\n")
```

Shapefile region names:

```
print(unique(ph_regions$shape1))
```

```
[1] "ARMM"           "CAR"           "NCR"
[4] "IlocosRegion"   "Cagayan Valley" "Central Luzon"
[7] "Calabarzon"     "Mimaropa"      "Zamboanga Peninsula"
[10] "BicolRegion"
```

```
# Read birth data (skip 2 lines to get past title/empty rows)
birth_data <- read_csv("PSA_both.t.csv", skip = 2)
```

Rows: 17 Columns: 21

— Column specification —

Delimiter: ","

chr (1): Region, Province/Highly Urbanized City

dbl (20): 2013 Male, 2013 Female, 2014 Male, 2014 Female, 2015 Male, 2015 Fe...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Calculate 2022 totals per region
birth_2022 <- birth_data %>%
  mutate(
    Total_2022 = `2022 Male` + `2022 Female`,
    Region = `Region, Province/Highly Urbanized City`
  ) %>%
  select(Region, Total_2022) %>%
  # Comprehensive region name cleaning to match shapefile
  mutate(
    Region_Clean = case_when(
      str_detect(Region, "NCR") ~ "NCR",
      str_detect(Region, "CAR") ~ "CAR",
      str_detect(Region, "ILOCOS") ~ "Ilocos Region",
      str_detect(Region, "CAGAYAN") ~ "Cagayan Valley",
      str_detect(Region, "CENTRAL LUZON") ~ "Central Luzon",
      str_detect(Region, "CALABARZON") ~ "Calabarzon",
      str_detect(Region, "MIMAROPA") ~ "Mimaropa",
      str_detect(Region, "BICOL") ~ "Bicol Region",
      str_detect(Region, "WESTERN VISAYAS") ~ "Western Visayas",
      str_detect(Region, "CENTRAL VISAYAS") ~ "Central Visayas",
      str_detect(Region, "EASTERN VISAYAS") ~ "Eastern Visayas",
      str_detect(Region, "ZAMBOANGA") ~ "Zamboanga Peninsula",
      str_detect(Region, "NORTHERN MINDANAO") ~ "Northern Mindanao",
      str_detect(Region, "DAVAO") ~ "Davao Region",
      str_detect(Region, "SOCCSKSARGEN") ~ "Soccsksargen",
      str_detect(Region, "CARAGA|Caraga") ~ "Caraga",
      str_detect(Region, "ARMM|BARMM") ~ "ARMM",
      TRUE ~ Region
    )
  )
```

```
# Print cleaned region names for debugging
cat("\nCleaned birth data region names:\n")
```

Cleaned birth data region names:

```
print(unique(birth_2022$Region_Clean))
```

```
[1] "NCR"           "CAR"           "Ilocos Region"
[4] "CagayanValley" "Central Luzon" "Calabarzon"
[7] "Mimaropa"      "Bicol Region"  "Western Visayas"
[10] "CentralVisayas" "Eastern Visayas" "Zamboanga Peninsula"
[13] "NorthernMindanao" "Davao Region" "Soccsksargen"
[16] "Caraga"        "ARMM"
```

```
# Perform join using cleaned region names
map_data <- ph_regions %>%
  left_join(birth_2022, by = c("shape1" = "Region_Clean"))

# Check join success
cat("\nJoin results - regions with missing birth data:\n")
```

Join results - regions with missing birth data:

```
missing_data <- map_data[is.na(map_data$Total_2022), "shape1", drop = TRUE]
print(missing_data)
```

```
character(0)
attr(,"class")
[1] "character"
```

```
cat("\nSummary of Total_2022 after join:\n")
```

Summary of Total_2022 after join:

```
print(summary(map_data$Total_2022))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
24820	43068	59824	91866	144340	216792

```
# Create static choropleth map using ggplot2
static_choropleth <- ggplot(map_data) +
  geom_sf(aes(fill = Total_2022), color = "white", size = 0.3) +
  scale_fill_gradient(
    name = "Live Births\n(2022)",
```

```

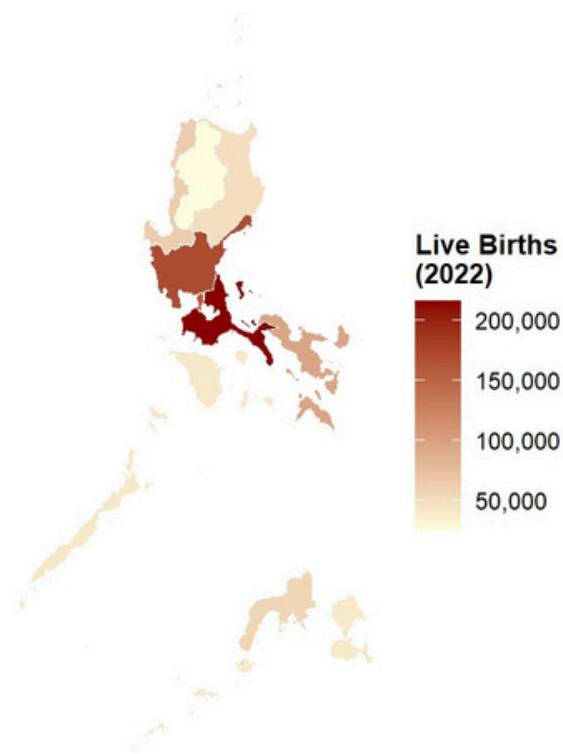
low = "lightyellow",
high = "darkred",
na.value = "lightgray",
labels = scales::comma_format()
}
labs(
  title = "Live Births by Region - Philippines 2022",
  subtitle = "Based on PSA OpenStat Data",
  caption = "Source: Philippine Statistics Authority (PSA)"
) +
theme_void() +
theme(
  plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
  plot.subtitle = element_text(hjust = 0.5, size = 12),
  plot.caption = element_text(hjust = 0.5, size = 8),
  legend.position = "right",
  legend.title = element_text(size = 10, face = "bold"),
  legend.text = element_text(size = 9)
)

# Display the static map
print(static_choropleth)

```

Live Births by Region - Philippines 2022

Based on PSA OpenStat Data



Source: Philippine Statistics Authority (PSA)


```
# Optional: Save the map
ggsave("philippines_births_2022_choropleth.png",
      plot = static_choropleth,
      width = 12, height = 8,
      dpi = 300,
      bg = "white")

# Print summary statistics
cat("\nTop 5 regions by live births in 2022:\n")
```

Top 5 regions by live births in 2022:

```
top_regions <- map_data %>%
  st_drop_geometry() %>%
  filter(!is.na(Total_2022)) %>%
  arrange(desc(Total_2022)) %>%
  select(shape1, Total_2022) %>%
  head(5)
print(top_regions)
```

```
shape1    Total_2022    1
Calabarzon    216792    2
Central Luzon 170318 3 NCR
160132    4    Bicol    Region
96962
5 Ilocos Region 63240
```

- Using PSA OpenStat (2013-2022) data and a Philippines regional shapefile, create an animated choropleth map showing how the number of live births per region changes over time. Color the regions based on **birth count**

```
# Read the shapefile
ph_regions <- st_read("philippines_Regions_level_1.shp")
```

```
Reading layer `philippines_Regions_level_1' from data source
`C:\Users\KImberly\R - Ylrebmik\Advanced Visualizations\philippines_Regions_level_1.shp'
using driver `ESRI Shapefile'
Simple feature collection with 10 features and 7 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box: xmin: 114.2779 ymin: 4.587294 xmax: 125.0077 ymax: 21.12189
Geodetic CRS: WGS 84
```

```
# Print region names in shapefile for debugging
cat("Shapefile region names:\n")
```

Shapefile region names:

```
print(unique(ph_regions$shape1))
```

```
[1] "ARMM"          "CAR"          "NCR"
[4] "IlocosRegion"  "Cagayan Valley" "Central Luzon"
[7] "Calabarzon"    "Mimaropa"      "Zamboanga Peninsula"
[10] "BicolRegion"
```

```
# Read birth data (skip 2 lines to get past title/empty rows)
birth_data <- read_csv("PSA_both.t.csv", skip = 2)
```

Rows: 17 Columns: 21

— Column specification —————

Delimiter: ","

chr (1): Region, Province/Highly Urbanized City

dbl (20): 2013 Male, 2013 Female, 2014 Male, 2014 Female, 2015 Male, 2015 Fe...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Calculate 2022 totals per region
birth_2022 <- birth_data %>%
  mutate(
    Total_2022 = `2022 Male` + `2022 Female`,
    Region = `Region, Province/Highly Urbanized City`
  ) %>%
  select(Region, Total_2022) %>%
  # Comprehensive region name cleaning to match shapefile
  mutate(
    Region_Clean = case_when(
      str_detect(Region, "NCR") ~ "NCR",
      str_detect(Region, "CAR") ~ "CAR",
      str_detect(Region, "ILOCOS") ~ "Ilocos Region",
      str_detect(Region, "CAGAYAN") ~ "Cagayan Valley",
      str_detect(Region, "CENTRAL LUZON") ~ "Central Luzon",
      str_detect(Region, "CALABARZON") ~ "Calabarzon",
      str_detect(Region, "MIMAROPA") ~ "Mimaropa",
      str_detect(Region, "BICOL") ~ "Bicol Region",
      str_detect(Region, "WESTERN VISAYAS") ~ "Western Visayas",
      str_detect(Region, "CENTRAL VISAYAS") ~ "Central Visayas",
      str_detect(Region, "EASTERN VISAYAS") ~ "Eastern Visayas",
      str_detect(Region, "ZAMBOANGA") ~ "Zamboanga Peninsula",
      str_detect(Region, "NORTHERN MINDANAO") ~ "Northern Mindanao",
      str_detect(Region, "DAVAO") ~ "Davao Region",
      str_detect(Region, "SOCCSKSARGEN") ~ "Soccsksargen",
      str_detect(Region, "CARAGA|Caraga") ~ "Caraga",
      str_detect(Region, "ARMM|BARMM") ~ "ARMM",
      TRUE ~ Region
    )
  )
```

```
# Print cleaned region names for debugging
cat("\nCleaned birth data region names:\n")
```

Cleaned birth data region names:

```
print(unique(birth_2022$Region_Clean))
```

```
[1] "NCR"           "CAR"           "Ilocos Region"
[4] "CagayanValley" "Central Luzon" "Calabarzon"
[7] "Mimaropa"      "Bicol Region"  "Western Visayas"
[10] "CentralVisayas" "Eastern Visayas" "Zamboanga Peninsula"
[13] "NorthernMindanao" "Davao Region" "Soccsksargen"
[16] "Caraga"        "ARMM"
```

```
# Perform join using cleaned region names
map_data <- ph_regions %>%
  left_join(birth_2022, by = c("shape1" = "Region_Clean"))

# Check join success
cat("\nJoin results - regions with missing birth data:\n")
```

Join results - regions with missing birth data:

```
missing_data <- map_data[is.na(map_data$Total_2022), "shape1", drop = TRUE]
print(missing_data)
```

```
character(0)
attr(,"class")
[1] "character"
```

```
cat("\nSummary of Total_2022 after join:\n")
```

Summary of Total_2022 after join:

```
print(summary(map_data$Total_2022))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
24820	43068	59824	91866	144340	216792

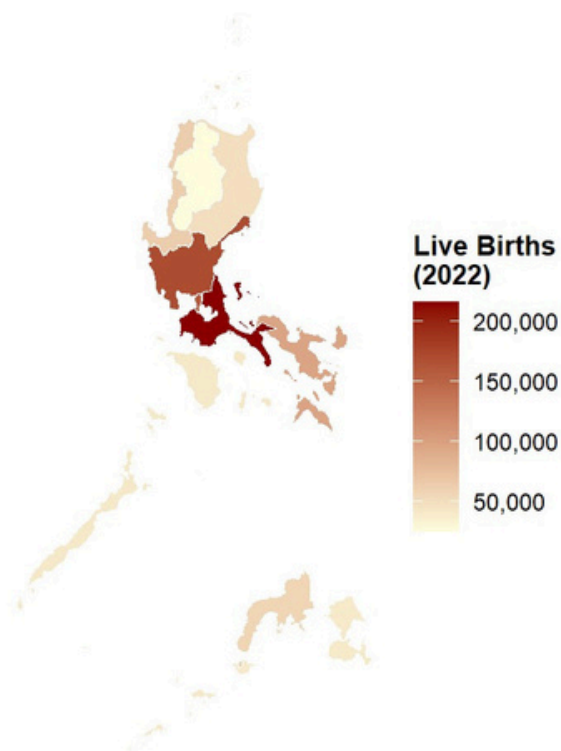
```
# Create static choropleth map using ggplot2
static_choropleth <- ggplot(map_data) +
  geom_sf(aes(fill = Total_2022), color = "white", size = 0.3) +
  scale_fill_gradient(
```

```
name = "Live Births\n(2022)",
low = "lightyellow",
high = "darkred",
na.value = "lightgray",
labels = scales::comma_format()
}
labs(
  title = "Live Births by Region - Philippines 2022",
  subtitle = "Based on PSA OpenStat Data",
  caption = "Source: Philippine Statistics Authority (PSA)"
) +
theme_void() +
theme(
  plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
  plot.subtitle = element_text(hjust = 0.5, size = 12),
  plot.caption = element_text(hjust = 0.5, size = 8),
  legend.position = "right",
  legend.title = element_text(size = 10, face = "bold"),
  legend.text = element_text(size = 9)
)

# Display the static map
print(static_choropleth)
```

Live Births by Region - Philippines 2022

Based on PSA OpenStat Data



Source: Philippine Statistics Authority (PSA)

```
# Optional: Save the map
ggsave("philippines_births_2022_choropleth.png",
      plot = static_choropleth,
      width = 12, height = 8,
      dpi = 300,
      bg = "white")

# Print summary statistics
cat("\nTop 5 regions by live births in 2022:\n")
```

Top 5 regions by live births in 2022:

```
top_regions <- map_data %>%
  st_drop_geometry() %>%
  filter(!is.na(Total_2022)) %>%
  arrange(desc(Total_2022)) %>%
  select(shape1, Total_2022) %>%
  head(5)
print(top_regions)
```

shape1	Total_2022	1
Calabarzon	216792	2
Central Luzon	170318	3
NCR	160132	4
Bicol Region	96962	5
Ilocos Region	63240	

How do birth trends differ across Philippine regions?

From 2013 to 2022, birth trends across Philippine regions reveal clear geographic and developmental differences. Urban areas such as NCR and CALABARZON consistently record the highest number of births, yet they also show the most significant declines. This is largely due to urbanization, higher levels of education, and the tendency to start families later. Luzon regions lead in total birth counts, while rural and more isolated areas like MIMAROPA and CAR display more fluctuating patterns but tend to have higher fertility rates. Although all regions are experiencing a nationwide shift toward lower birth rates, the speed of this transition differs depending on the region's economic status, degree of urbanization, and availability of family planning services.

Which regions have experienced the highest increase or decrease in live births from 2013 to 2022?

From 2013 to 2022, most regions in the Philippines saw a decline in live births, with the sharpest reductions likely occurring in highly urbanized areas such as NCR and CALABARZON. These trends can be attributed to economic challenges, a greater focus on careers, and improved access to family planning. In contrast, rural regions like CAR and the Ilocos Region may have experienced smaller decreases in absolute numbers but still aligned with the overall national downward trend. The decline likely intensified during 2020 to 2022, as the COVID-19 pandemic brought about economic instability, disruptions in healthcare services, and delays

in family planning, with urban areas possibly feeling these effects more strongly due to stricter lockdowns and greater economic uncertainty.

How does using `gganimate` with `sf` enhance your ability to analyze spatial(and time) data trends?

Combining `gganimate` with `sf` transforms static geographic analysis into a dynamic spatiotemporal visualization by utilizing the human ability to perceive motion and changes over time. This technique allows users to easily identify trends, outliers, and key turning points without the need to compare multiple static maps. It also reveals patterns of geographic clustering and shows how regions either converge or diverge as time progresses. Animated visualizations improve analytical understanding by making correlations and recurring patterns more visible, while also enhancing communication by helping stakeholders quickly interpret complex demographic changes and make informed policy decisions based on clear spatial and temporal trends.